

7th Heaven Cafe: Business Logic Design

Core Entities and Data Modeling

The system's data architecture is built on interconnected entities that track the lifecycle of an artisanal bakehouse and cafe's digital operations, customer orders, and kitchen workflows.

- **User (Admin & Customer):** Represents cafe administrators, staff, or registered customers. It stores profile credentials, full name, email, phone number, saved delivery addresses, and role-based permissions (Customer, Kitchen, Delivery, Admin). Security logic mandates that all user sessions are verified using secure Firebase Authentication tokens.
- **MenuItem:** Represents an artisanal food or beverage offering (Hot Coffee, Cold Coffee, Refreshers, Milkshakes, Cupcake Milkshakes, Mug Cakes). It includes attributes such as name, category, selling price, detailed description, preparation time, and high-resolution media URLs. An `inStock` boolean field allows kitchen staff to instantly toggle availability without deleting the catalog record.
- **Order:** Represents a customer purchase transaction. It is linked to the user account (or flagged as a verified guest order) and encapsulates delivery address, payment method (Online Razorpay or Cash on Delivery), payment status, fulfillment status, timestamp, and estimated preparation/delivery time.
- **OrderItem:** Operates as a point-in-time snapshot embedded within an Order document. It records the exact item name, unit price, quantity, and selected customization addons at the precise moment of purchase to preserve historical financial integrity, regardless of subsequent menu pricing changes.
- **OrderSequence:** A synchronized sequence counter used to generate customer-facing, formatted sequential invoice numbers (e.g., 7H-20260303-0042) to ensure human-readable tracking across kitchen slips and email invoices.

Authentication and Role-Based Access

Cafe operations are segregated into distinct access levels to eliminate friction for online ordering while safeguarding kitchen management and sensitive sales data.

- **Admin & Kitchen Access:** Cafe managers and kitchen staff authenticate with administrative credentials. Administrative authorization unlocks protected interfaces, including real-time Kanban kitchen queues, item stock toggles, live revenue analytics charts, and menu catalog management.
- **Customer & Guest Access:** Customers interact with the web platform through seamless browsing. Customers can either place orders as guest users or sign in with Google/Email to retain saved delivery addresses, view comprehensive order histories, and access 1-click reordering. No mandatory app installation is required.

Order Processing and Pricing Logic

The checkout workflow enforces strict server-side validation rules to prevent price tampering, guarantee inventory availability, and ensure financial accuracy.

- **Server-Side Price Calculation:** When a customer submits a cart for checkout, the system independently fetches current catalog records from the database. The subtotal, 5% GST tax, packaging/delivery fees (waived on orders exceeding ₹500), and coupon deductions are computed server-side, actively discarding any untrusted totals submitted by the browser.
- **Availability Verification:** The system verifies that each selected item is flagged as active and in-stock. If an item is sold out, the transaction is gracefully halted and the customer is notified before payment initiation.
- **Payment Processing & Verification:** For online transactions, Razorpay order IDs are created server-side via `/api/razorpay`. After payment authorization, cryptographically verified signatures guarantee transaction completion before the order transitions to the active kitchen queue.

Kitchen Queue and Lifecycle Management

Order fulfillment follows a deterministic state machine, prioritizing kitchen throughput, order accuracy, and real-time customer visibility.

- **First-Come-First-Served (FCFS) Real-Time Queue:** The kitchen display dashboard utilizes real-time Firestore listeners (onSnapshot) to receive incoming orders instantly. Active tickets are chronologically sorted to enforce an FCFS preparation sequence.
- **Status Progression:** Kitchen staff and delivery personnel transition tickets through four discrete statuses: **Placed** (order verified and queued), **Preparing** (kitchen actively brewing/baking with estimated countdown), **Out for Delivery** (courier dispatched with customer contact), and **Delivered** (handoff complete).
- **Customer Real-Time Tracking:** The customer tracking view subscribes directly to their unique order document. Status badges, preparation countdown timers, and courier progress update reactively in real time without requiring manual page reloads.
- **Data Aggregation & Post-Fulfillment:** Upon reaching the Delivered state, the system automatically triggers an itemized HTML invoice email via Nodemailer and prompts the customer for an optional star rating and review. Completed transactions aggregate into daily revenue metrics for executive reporting.

7th Heaven Cafe: Database Design

This document outlines the database design for the 7th Heaven Cafe Web Platform, implemented using Google Cloud Firestore (NoSQL) and the Firebase Client/Admin SDKs. The architecture leverages document-based storage, embedding arrays for point-in-time snapshots (order items and customized addons) while maintaining clean document relationships for users, menu items, and real-time orders.

1. User Collection

Stores registered cafe customers, administrative staff, and delivery accounts.

Field Name	Data Type	Constraints / Description
uid	String	Primary Key. Matches Firebase Authentication UID.
name	String	Required. Customer or staff full name.
email	String	Required, Unique. Email address for login and invoice delivery.
phone	String	Optional. Contact number for delivery coordination.
role	String	Required. Enum: ['customer', 'kitchen', 'delivery', 'admin']. Default: 'customer'.
photo	String	Optional. Profile image avatar URL.
addresses	Array of Objects	List of saved delivery addresses (tag, street, city, pincode, isDefault).
newsletter	Boolean	Default: true. Customer marketing email preference.
createdAt	Timestamp / ISO	Timestamp when the account record was created.
updatedAt	Timestamp / ISO	Timestamp when the profile was last modified.

2. MenuItem Collection

Stores individual food and beverage products offered by 7th Heaven Cafe.

Field Name	Data Type	Constraints / Description
id	String	Primary Key. Unique Firestore auto-generated document ID.
name	String	Required. Name of the menu item (e.g., 'Caramel Flan Latte').
category	String	Required. Enum: ['Hot Coffee', 'Cold Coffee', 'Refreshers', 'Milkshake', 'Cupcake Milkshake', 'Mug Cakes'].

Field Name	Data Type	Constraints / Description
price	Number	Required. Current base selling price in INR (₹).
description	String	Required. Culinary description and flavor profile of the item.
image	String	Required. Cloud Storage URL for the item photograph.
inStock	Boolean	Default: true. Kitchen stock toggle to show or hide from customer menu.
isVegetarian	Boolean	Default: true. Dietary indicator badge.
isBestseller	Boolean	Default: false. Promotes item to featured recommendations.
prepTimeMinutes	Number	Estimated standard preparation time in minutes (default: 5-10 mins).
addons	Array of Objects	Optional customizable additions (e.g., Extra Espresso Shot, Oat Milk).
createdAt	Timestamp / ISO	Timestamp when the item was added to the catalog.

3. Order Collection

Records all customer orders, encapsulating a point-in-time snapshot of purchased items, financial breakdown, and live lifecycle status.

Field Name	Data Type	Constraints / Description
id	String	Primary Key. Unique Firestore auto-generated document ID.
orderNumber	String	Formatted sequential order identifier (e.g., '7H-20260303-0042').
userId	String	Required. User UID if authenticated, or 'guest' for guest checkouts.
customerName	String	Required. Full name provided during checkout.
customerEmail	String	Required. Recipient email address for digital invoice dispatch.
customerPhone	String	Required. Customer contact number for delivery coordination.
customerAddress	String	Required for delivery orders. Street, landmark, city, and pincode.
orderType	String	Enum: ['delivery', 'dine_in', 'takeaway']. Default: 'delivery'.
items	Array of Objects	Required. Embedded sub-documents containing historical snapshots of items ordered.
items.menuItemId	String	Reference to MenuItem document ID.
items.name	String	Required. Frozen snapshot of item name at time of purchase.
items.price	Number	Required. Frozen snapshot of unit price at time of purchase.
items.qty	Number	Required. Number of units ordered.
items.selectedAddons	Array of Objects	Optional addons snapshot (name and addon price).
items.itemTotal	Number	Calculated line total: (price + sum of addons) * quantity.
subtotal	Number	Sum of item totals before taxes and delivery.
taxes	Number	Calculated 5% GST on food & beverage items.
deliveryFee	Number	Packaging & delivery fee (₹40, or ₹0 if subtotal ≥ ₹500).
discount	Number	Promotional coupon discount deduction.

Field Name	Data Type	Constraints / Description
grandTotal	Number	Required. Final net amount: subtotal + taxes + deliveryFee - discount.
paymentMethod	String	Required. Enum: ['Online (Razorpay)', 'Cash on Delivery'].
paymentStatus	String	Required. Enum: ['pending', 'paid', 'failed', 'refunded'].
paymentId	String	Optional. Razorpay transaction ID (e.g., 'pay_PAB91823749').
status	String	Required. Enum: ['placed', 'preparing', 'out_for_delivery', 'delivered', 'cancelled'].
placedAt	Timestamp / ISO	Timestamp of when the customer completed checkout.
estimatedDeliveryAt	Timestamp / ISO	Calculated delivery ETA for customer live countdown (default: +25 mins).
deliveredAt	Timestamp / ISO	Timestamp when courier marks order as successfully delivered.
rating	Number	Optional. Post-delivery customer rating (1 to 5 stars).
feedback	String	Optional. Customer written review or comment.

4. Coupon Collection

Stores promotional discount codes, validity windows, and discount rules.

Field Name	Data Type	Constraints / Description
code	String	Primary Key. Uppercase alphanumeric code (e.g., 'HEAVEN20').
discountType	String	Required. Enum: ['percentage', 'flat'].
discountValue	Number	Required. Percentage amount (e.g., 20) or flat rupee amount.
maxDiscount	Number	Optional. Upper ceiling limit in INR for percentage discounts.
minOrderValue	Number	Required minimum subtotal threshold to apply coupon.
validUntil	Timestamp / ISO	Expiration timestamp after which the coupon is rejected.
isActive	Boolean	Default: true. Admin toggle to enable or deactivate campaign.

Relationships & Data Integrity Rules

- Snapshot Pattern (OrderItems):** Line items within an order are embedded as frozen objects rather than referenced via dynamic foreign keys. The item name, unit price, quantity, and selected addons are captured at the moment of checkout. This guarantees that future price updates or catalog deletions by the cafe administrator will never alter historical financial transactions or customer invoices.
- Role & Multi-Tenant Isolation:** Firestore Security Rules enforce strict boundaries: customers can only read and write their own profile and order records (`auth.uid == userId` or guest order ID match), while administrative staff possess elevated privileges for status progression, catalog editing, and sales aggregation.
- Real-Time Synchronization:** Real-time event listeners (`onSnapshot`) connect the customer order tracking view and the kitchen display board directly to Firestore document channels. Status transitions (`placed` → `preparing` → `out_for_delivery` → `delivered`) propagate instantaneously without polling overhead.